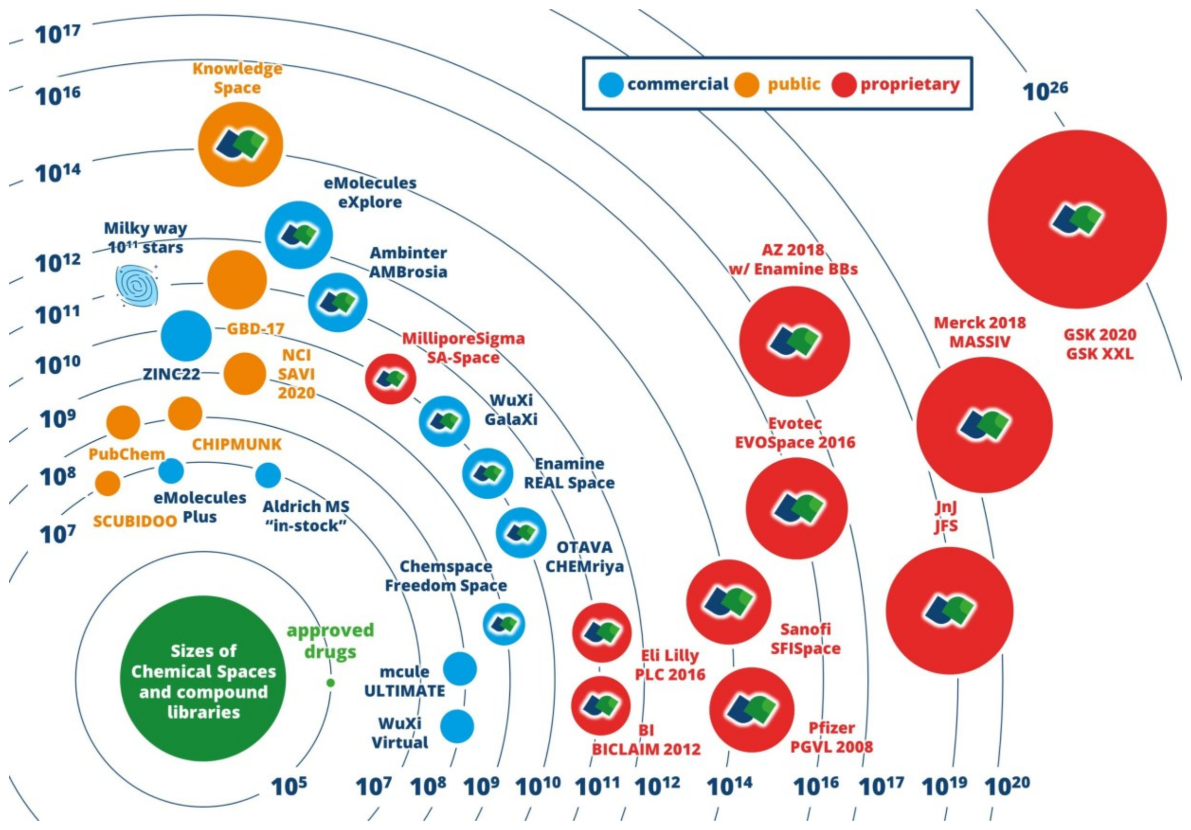
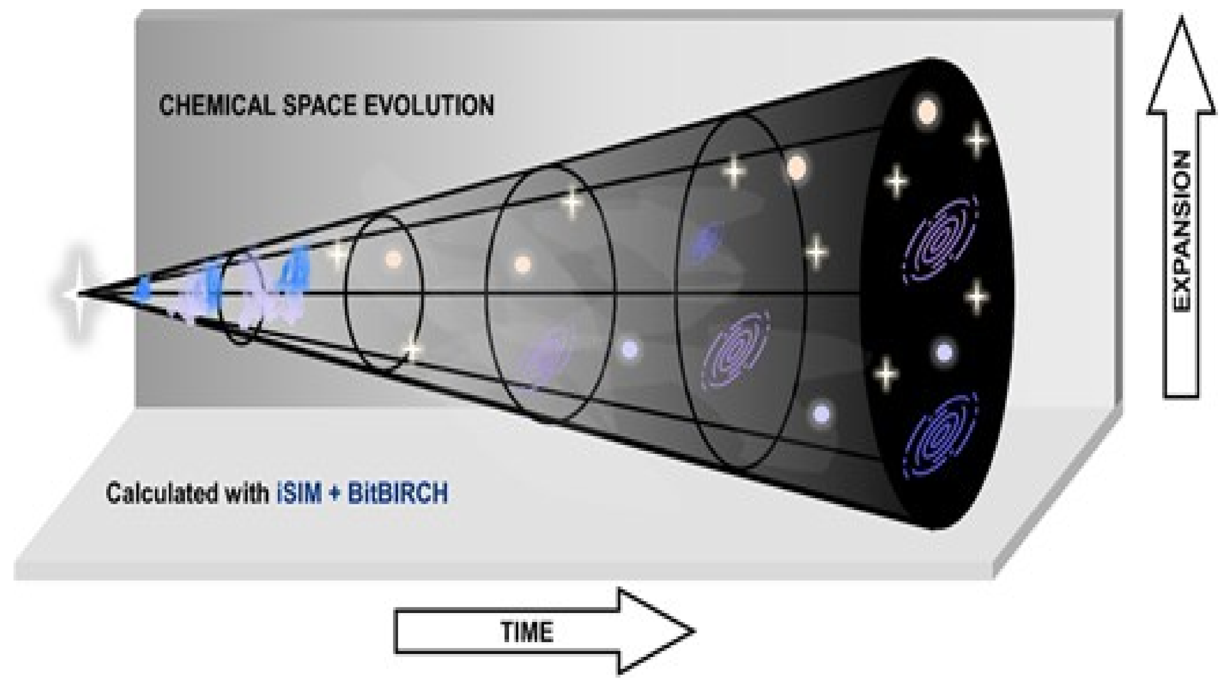
Mapping Biologically Relevant Chemical Space: The Power of Chemoinformatics and AI

Chemical space:
The set of all possible molecules. Each molecule has different coordinates based on its properties.

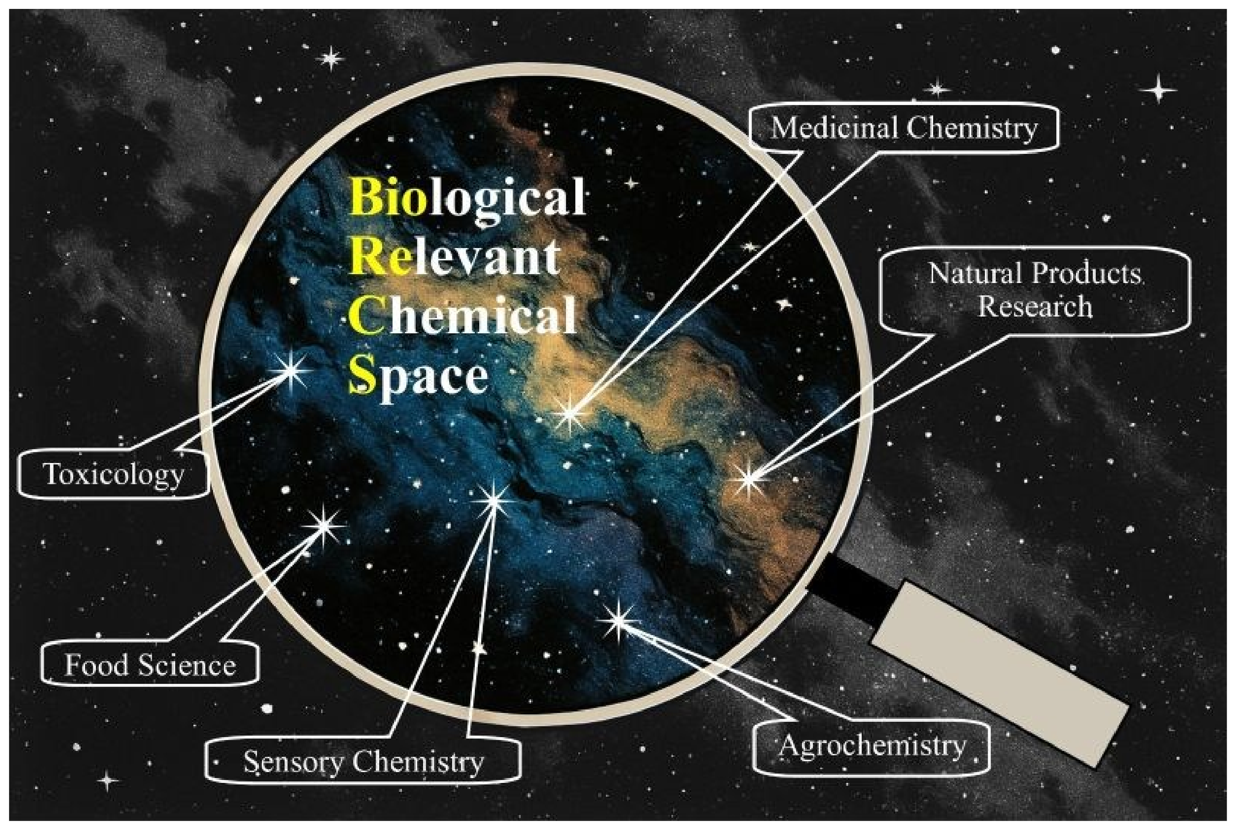
Scale of the Chemical Universe:
It is estimated that the number of molecules amounts to 10^{60}, but only 10^{25} have been explored..



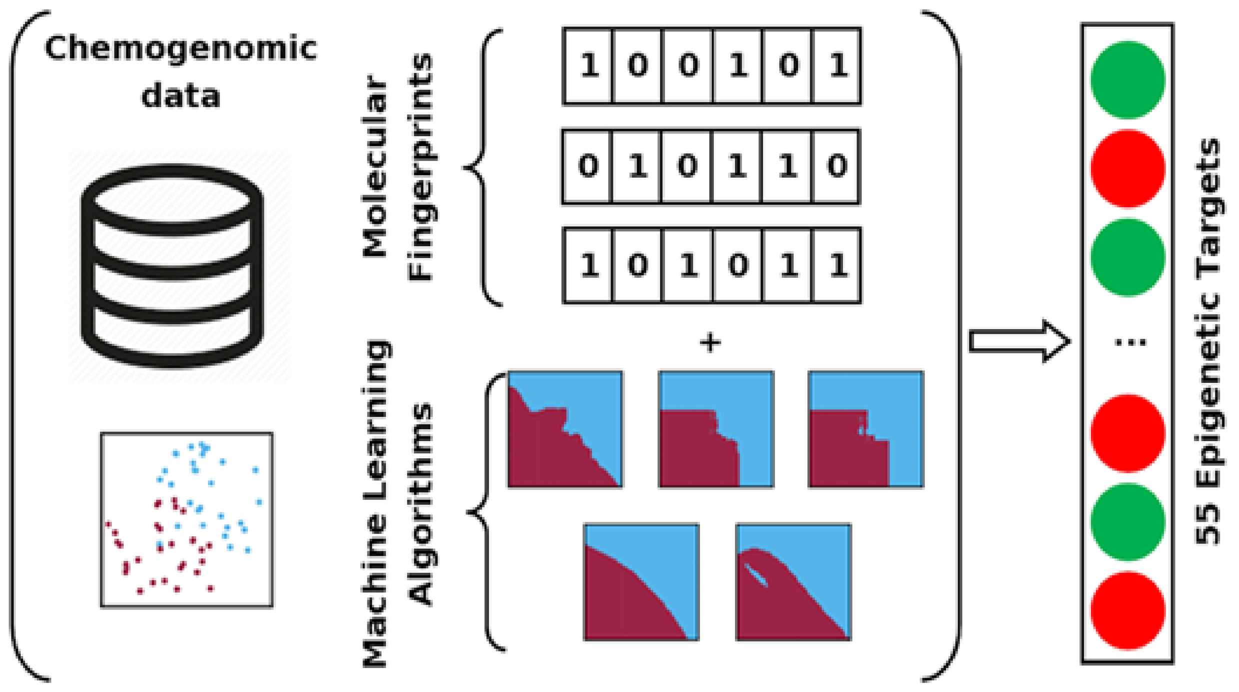
Growth vs Diversity of the Chemical Space:
An increasing number of molecules cannot be directly translated to diversity.



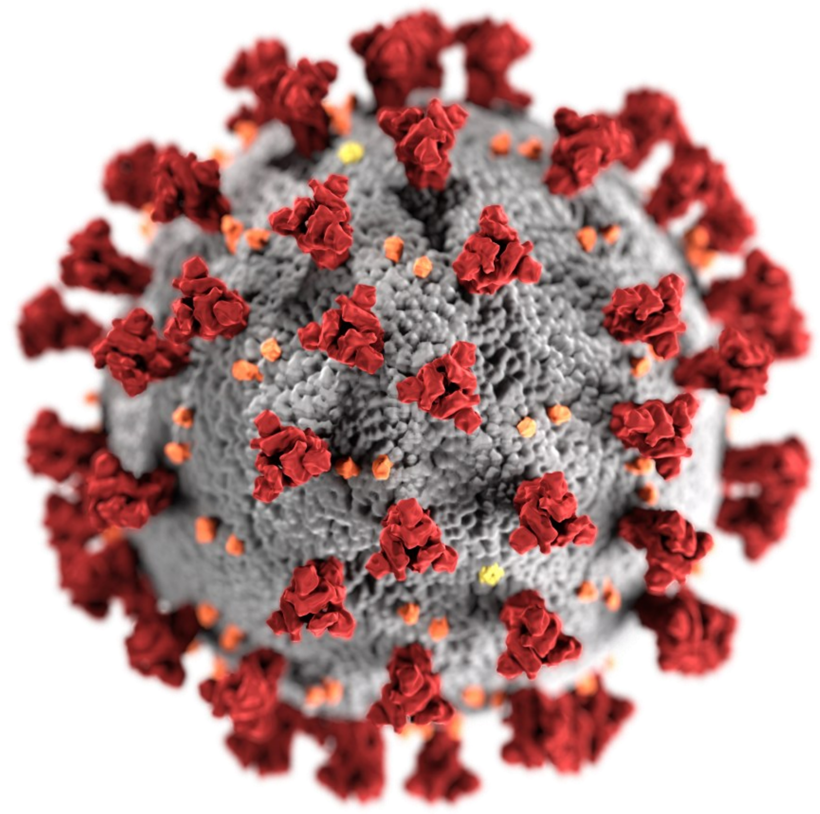
Biologically Relevant Chemical Space:
Specific subset containing bioactive molecules.



Case Study 1: Discovery and development of DNMT inhibitors.
260000 starting compounds, 24 in silico hits, 15 tested, 7 experimental hits.



Case Study 2: Identification of SARS-CoV-2 Main Protease Inhibitors.
17 M starting compounds, 18 in silico hits, 12 tested, 1 experimental hits (+3 similar).



On the pursuit of the true impact of our actions in science:
Beyond grades and impact factors, what matters is purpose and a well-rounded education.

Conclusions:

- Chemoinformatics and AI optimize the search for HITs, but their experimental validation remains essential for validating the models.
- In the vastness of theoretical chemical space (most of which remains unexplored), we can find regions containing molecules of biological interest; however, an increasing number of molecules cannot be directly translated into diversity.
- Metrics are not the end goal but rather the means to our scientific endeavors.